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An investigation of the effects of game difficulty on player enjoyment

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ABSTRACT

Motivated by the controversial debates on the subject of static difficulty and dynamic difficulty adjustment (DDA), we investigate the effects different difficulty settings have on different types of players (i.e. casual or experienced). By doing this, we can help to deduce an important question: what difficulty does the player want? An experiment was conducted with 90 participants, each playing the same experimental game exhibiting either static difficulty or DDA mechanics. The results indicate that players enjoy a game more, if the difficulty provided is reflective of their gaming experience, rather than their actual gaming ability.
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1. Introduction

"Once upon a time, games were competitors. Now, primarily, they're entertainers. They aimed to beat you. Now, to be beaten" [1].

The video game industry has evolved considerably, introducing a whole new set of consumers. The Nintendo Wii especially, piloted this change by providing a more accessible experience to the casual market. Consequently, different difficulty methods have been imposed in a variety of games to help ensure that both casual and hardcore players with varying skill levels stay engaged throughout their gaming sessions. *Mario Kart Wii* for instance, bridged the gap between the two audiences via its rubber-banding mechanics and the *Wii Wheel*. A common issue illustrated when creating a game for different audiences varying in ability, is that a game too hard causes frustration, whereas a game too easy causes boredom. The aim of the game designer therefore is to help the player reach their "sweet spot", by balancing the effects produced by poor difficulty management; commonly referred to as Flow [2].

Generally there are two approaches used to resolve this problem. Either to provide the player static difficulty options to choose from (i.e. easy, medium or hard) or to adapt the game's difficulty to the player's ability using dynamic difficulty adjustment (DDA) methods. The subject of the "correct" difficulty approach to use has sparked controversy between game designers and writers alike [3–6].

Commonly, DDA is referred to as the system to prevent the player from becoming bored from finding the game too easy, or becoming too frustrated because of the game's high difficulty [5,3]. However, Passive DDA disregards the fact that not every individual wants to be challenged at the highest of their ability continuously; in fact, by doing so the player may "become physically and/or mentally tired after sustained play" [7]. They may instead wish to play at their own pace [8], because each individual has their own "physiological motivators" for playing any game [9]. Why the player wishes to play (whether for challenge or enjoyment), in conjunction with the user's perceived ability establishes their difficulty choice.

Atkinson [10] proposed that negatively motivated individuals prefer to choose tasks that are very easy or very complex, whilst the positively motivated, tend to choose more realistic, yet moderately difficult tasks. In terms of computer games, a casual player tends to look at games as a form of relaxation and pleasure" [11], suggesting that they are more likely to choose an easier difficulty, thus demonstrating negative motives. Contrarily, an experienced player models the characteristics of someone who has a higher motive to achieve success, since they continuously practice [11] and ultimately play to win, portraying positive motives. Obviously there are other factors that contribute to the difficulty chosen, such as the targeted audience for the game being played. A casual player for instance, would feel more comfortable playing *Farmville* as opposed to *Super Meatboy*.

Kruger and Dunning [12] believed that unskilled performers overestimate their own performance greatly, in comparison with

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skilled performers who tend to underestimate their performance somewhat. Unskilled performers, who were later trained however, were able to be more accurate in their self-assessment. This demonstrates that regardless of ability – whether skilled or unskilled, experience is required in order to have good self-assessment.

Motivated by the controversial debates over game difficulty management systems [3–6] and the different models designed to resolve this debate [8,13], we investigated the effects of game difficulty on player behaviour. In particular, the behaviour of casual and experienced players will be analysed, when playing an experimental game utilising static difficulty and DDA mechanics.

By analysing the behaviour of different types of players, it helps us to discover *what difficulty does the player actually want?* We firstly present our predicted outcomes based on literature we reviewed and then introduce Star-Surf; our experiment. This is followed by an analysis of the results gathered from the test. Finally, we conclude our findings and provide opportunities for future work.

2. Hypotheses

The following hypotheses were proposed in consideration of the literature we reviewed, to help us discover trends between different types of players and the reasoning behind these trends.

Hypothesis 1. Experienced players will demonstrate better self-assessment

With consideration of Dunning and Kruger's study [12], experienced players should demonstrate the best self-assessment, since they are more likely to be skilled and trained. Therefore they should have better meta-cognitive ability, than casual players. Some casual players will likely be skilled, whilst others will not; therefore the poorest performers in the casual group should demonstrate the poorest self-assessment.

Hypothesis 2. Casual players will choose easier difficulties regardless of their ability

Following Atkinson's achievement of motivation [10], casual players will likely select difficulties that are relatively easy regardless of their actual ability, since the casual player profile typically portrays negative motives. Experienced players however, will choose more moderate difficulties, such as medium or hard, since they enjoy the challenge.

Hypothesis 3. Casual gamers will enjoy easier difficulty levels, but experienced gamers will not

Gilleade, Dix, and Allanson [9] demonstrated that casual players became frustrated much more easily than experienced players when the game's difficulty increased. Therefore the casual players who pursue easier difficulties will enjoy the game much more, than casual players who pursue harder difficulties. On the other hand, experienced players will not enjoy easier difficulties if it does not match their ability, since they desire to be challenged.

3. Creating Star-Surf

We designed Star-Surf to ensure that both casual and experienced players could naturally adapt to the game. It was developed with ActionScript 3.0 to target the Adobe Flash platform. To monitor the effects caused by different difficulty scenarios, the same game is used; however the participant has a random chance of playing the game in static difficulty mode or DDA mode.

3.1. The game

In Star-Surf the player takes control of a green square with their mouse cursor and must collect as many orbs/points as possible within a two minute time limit, by moving through red gates. The orbs and gates move from right to left continuously, producing the illusion that the player is moving. The Y position of each orb is chosen randomly and is surrounded by a gate. The size of the gate and the distance between each orb is influenced by the current difficulty level (Fig. 1).

3.1.1. Tutorial

When the player first starts the game, they are requested to select their preferred difficulty by using a slider. The slider uses float values from 0 to 1; where the left of the bar – 0, represents the easiest difficulty and the right side – 1, represents the hardest difficulty. To aid the player's judgement preset static difficulty options appear above the slider, depending on the current float value.

- Very Easy: 0 – 0.2
- Easy: 0.2 – 0.4
- Normal: 0.4 – 0.6
- Hard: 0.6 – 0.8
- Very Hard: 0.8 – 1

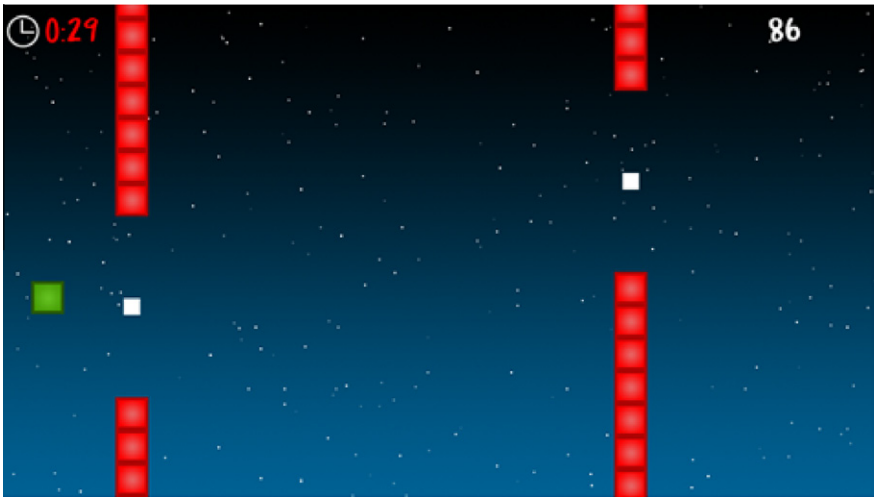


Fig. 1. A screenshot of Star-Surf.

Table 1
Layout of the tutorial procedure.

Text displayed	Procedure
Move using the mouse.	• After the player moves the mouse, a hidden five second countdown begins • On completion, the tutorial proceeds to the next stage
Collect upcoming orbs	• A total of five orbs scroll from right to left on the screen continuously • Once all orbs are collected the tutorial continues
... and avoid red obstacles	• A hidden five second countdown commences • On completion, the actual game begins

The default difficulty shown from the beginning is normal – 0.5. Once the player has selected their preferred difficulty and clicks the "PLAY GAME" button, the player is given a brief tutorial to ensure that they are aware of the game rules before being fully tested. At this stage a random difficulty method is also chosen for them. A set of text boxes appear on the screen to inform the player about the rules of the game (Table 1).

The player's performance in the tutorial does not affect their results or the game's difficulty. The player is also not informed about the difficulty method, to ensure that they are not influenced by knowledge of the current difficulty system.

3.1.2. In game

In static difficulty mode, the difficulty level gradually increases throughout the gameplay session. The difference between the starting difficulty and ending difficulty (i.e. the difficulty after the two minute time limit) is 20%. This has been done to imitate games with static difficulty; for example platformers typically have a number of levels that progressively increase in challenge. If the difficulty level reaches the hardest level possible, the difficulty level is clamped.

In DDA mode, the difficulty dynamically changes based on the player's performance. The difficulty level increases or decreases in the following situations:

1. If the character touches an orb, the difficulty increases by 0.01.
2. If an orb is missed by the character, the difficulty decreases by 0.01.
3. If the character collides with a post, the difficulty is decreased by 0.1.

These values have been used to ensure that the increase or decrease in difficulty is gradual. If the difficulty increased by a huge amount quickly, the new difficulty may be too hard for the player; therefore by gradually changing the difficulty, a more accurate difficulty matching the player's ability can be calculated.

3.1.2.1. Game difficulty. There are four noticeable parameters in the game that can be used to increase or decrease the game's level of difficulty as seen in Fig. 2.

1. The speed the posts move at
2. The height of the gate which has the orb
3. The width between each set of posts
4. The height between each gate from post to post

The gate size and distance between each post enlarges if the difficulty becomes easier. The speed also slows down if the difficulty level reduces. The opposite occurs in each of these cases, if the difficulty level increases.

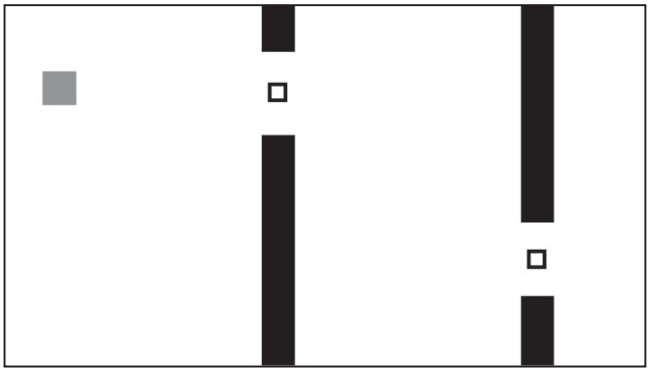


Fig. 2. Different difficulty parameters.

The first three parameters mentioned have been used and the fourth has not. The fourth difficulty parameter was not used, to prevent the positioning of the orbs becoming more predictable. If the difficulty level was currently easy, the orb Y locations within each gate would be quite similar. If the difficulty level was harder however, the Y positions of the orbs would likely alternate from the top of the screen to the bottom each time.

A mutual difficulty level is used to determine the values for each parameter. The reason the difficulties are set this way, rather than having an exclusive difficulty level for each parameter, is because it is difficult to determine when a specific parameter should be increased or decreased in difficulty.

For example, if the character collides with an obstacle there are a variety of potential causes:

1. The game is moving too fast for the player.
2. The gate size is not large enough for the player.
3. The width between each post is not big enough.

Since there are multiple reasons as to why the character could hit the wall, it is impossible to pick a particular option to identify one parameter that should be decreased in difficulty. Therefore, all parameters are updated according to a global difficulty value.

The following equation is used to calculate the speed, taking the current difficulty value into account:

$$\text{Difference} = \text{MaximumSpeed} - \text{MinimumSpeed};$$
$$\text{Speed} = (\text{difference} * \text{difficulty}) + \text{MinimumSpeed};$$
Equation 1 : Calculates the speed

To calculate the gate size and the width between each post, the following equation is used:

$$\text{Difference} = \text{Maximum} - \text{Minimum};$$
$$\text{value} = (\text{Maximum} - (\text{difference} * \text{difficulty}));$$
Equation 2 : Calculates the gate size and distance between each post

Obviously each of the three difficulty parameters needs to have a minimum and maximum value to ensure that the game is not impossible to play. For example, if the difficulty continuously increases, the gate size will decrease until the gate is too small to fit through. Likewise, if the difficulty continuously decreased, the speed of the game would eventually reach a halt at zero, without the use of a minimum value. Table 2 shows the different values used for each difficulty parameter.

The minimum speed is 2, to ensure that there is at least one orb and gate present on the screen at a time. The maximum speed has been set to 15 to ensure that it is possible for the player to traverse from one gate to the next.

The character's height and width is 32. The minimum gate size of 96 is three times the height of the character, which makes it appear quite simple to fit through each gate. This value is used to give the player a little room for error. If the minimum gate size used was 32, the player would be required to position the character exactly in the middle of the gate. It would be quite ridiculous to request the player to move to this exact position at the hardest difficulty every time. Arguably the value of 64 could be used; however through extensive play testing this value seemed ruthless and did not blend well with the minimum values used for the speed and the distance between each set of posts.

The maximum gate size value is the height of the screen, which gives the player minimal chance of colliding with a post. The minimum and maximum values used for the distance between two sets of obstacles were continuously changed throughout development, after countless play-test sessions. The minimum value ensures at least one orb is visible at a time and the maximum value complimented the other difficulty parameters nicely.

Hopefully the explanations of why these minimum and maximum values were used clarify how the equations mentioned previously, equate to the range of difficulty settings from very easy to very hard. It was important to address the understanding that went behind the range of difficulties, to demonstrate that the choices were not made by the designer's idea of these difficulties, but more so by the extreme limits of the game, described previously as minimum and maximum.

3.1.3. Post game

Once the two minute time limit has ended, a postgame screen is displayed, showing the player how many orbs they collected and the best combo they acquired (i.e. the highest amount of orbs they collected in a row, without hitting an obstacle or missing an orb).

When the player clicks on a "CONTINUE" button they are prompted with the following questions:

Are you a casual or experienced gamer? This has been asked to directly compare the results of casual players to experienced players, where a correlation between the two will hopefully be noticeable.

Did the game match your skill? This helps to identify if the player felt they chose the right difficulty (if in static difficulty mode) or if the dynamic difficulty implementation worked for them (if in DDA mode).

How much did you enjoy the game? This question allows us to observe correlations between player performance and their enjoyment. If the player performed well, they supposedly may give the game a high rating. This question is answered by selecting a point on a slider similar to how a user selects their preferred difficulty. The enjoyment values range from 1 (not enjoyable) to 10 (very fun).

How difficult did you find the game? This question is answered with the use of a slider between the scales of 1 (too easy) to 10 (too hard). This can also help to observe correlations between how hard the player found the game, to how much they enjoyed it.

How well do you think you did? Again, this question is answered via a slider between the scales of 1 (very bad) and 10 (awesome). This is used to record the player's self-assessment, which will then be compared to a difficulty they chose before the start of the game and their actual performance.

3.2. Participants

A URL to the location of Star-Surf was advertised on social networking websites. The participants that took part in the experiment were people within these communities, or people who were referred by these members. Details such as each participant's gender or age were not recorded. The important factor was discovering the level of gaming experience each of these participants had. There were 90 participants in total, consisting of 71 experienced players and 19 casual players.

3.3. Data collection

In DDA mode throughout each participant's gameplay session, the current difficulty was recorded every second. The game's two minute time limit allowed for an easier comparison to be made between different game sessions. By using the same time limit for each session, the difficulty levels of each player at different stages could be compared directly.

The answers to the questions mentioned previously were also recorded, in addition to:

- The player's final score
- The amount of times the player missed an orb
- The amount of times the player hit an obstacle

The performance of each participant varied depending on the difficulty mode. Since each participant could choose their own difficulty, it was difficult to compare each participant's performance in relation to other participants who have chosen different difficulty levels. The performance for each participant in static difficulty mode is the percentage of how many orbs they collected out of how many the participant could have.

The results obtained from DDA mode proved to be better at comparing the performance of each individual, since the game's difficulty adapted to the ability of the player. The two minute time limit ensured that the player had enough time to reach their peak performance, even if they chose a difficulty lower than their actual ability. The difficulty levels in the last 30 s of each participant's session are averaged to represent their actual performance.

4. Results and discussion

Following the three hypotheses made in Section 2, this section has divided the results collected and analysis made, into three subsections respectively:

1. *Player Performance and Self-Assessment*: This sub-section analyses the performance of the participants, to identify trends between the difficulty they chose and their perceived performance.
2. *Comparison of Skill and Ability:* This sub-section determines whether the players thought that the game's difficulty matched their ability.
3. *Enjoyment:* This sub-section analyses the enjoyment levels of casual and experienced players in both difficulty modes.

Throughout this section two different methods are used to calculate the actual performance of each participant:

1. *Method A:* Calculates the percentage of orbs collected to the total amount of orbs the player could have collected.
2. *Method B:* Calculates the mean of the difficulty levels traversed through in the last thirty seconds of their session (specific to DDA mode).

Method A allowed us to sort participants by their performance relative to their own gameplay session; since the amount of orbs that can be collected is influenced by the difficulty chosen (i.e. an increase in difficulty means an increase in speed, meaning more orbs will appear). Method B on the other hand, was more helpful for sorting participants in order of performance relative to every-

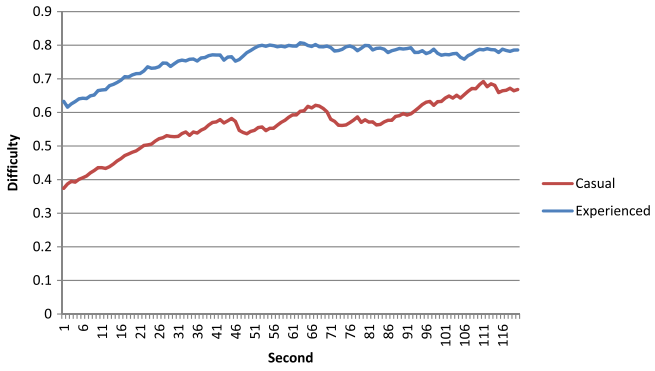


Fig. 3. The difficulty levels traversed through in DDA mode.

one else's, since in DDA mode the difficulty adjusts to the skill of the player.

4.1. Player performance and self-assessment

Fig. 3 represents the performance of an average casual ($n = 12$) and experienced ($n = 32$) player in DDA mode, throughout their two minute gameplay session. For each second, the mean difficulty level for both groups of players has been calculated.

Fig. 3 clearly demonstrates that each player (whether casual or experienced), chose a difficulty level that was a lot lower than their actual ability. This is very noticeable from observing the chosen difficulty (from the first second) to the peak difficulty level the player reached throughout their session. Predictably, experienced players performed better on average.

There is a very noticeable growth in difficulty throughout both gameplay sessions, with minor decays where players hit obstacles in the game. However, it took a longer time for casual players to reach their peak difficulty, due to the initial difficulty they chose; demonstrating that experienced players chose a difficulty more suited to their ability, than casual players.

Fig. 4 observes the perceived performance and actual performance of casual ($n = 19$) and experienced ($n = 71$) players, in both difficulty modes. The actual performance of each participant was calculated using Method A.

Fig. 4 magnifies the contrast in the difficulties chosen between casual and experienced players (as noticed in Fig. 3). The mean difficulty chosen by casual players was 0.37, whilst the mean difficulty chosen by experienced players was 0.6. This is interesting since casual players chose a difficulty slightly easier than the recommended difficulty (medium) whilst experienced players chose a difficulty slightly harder.

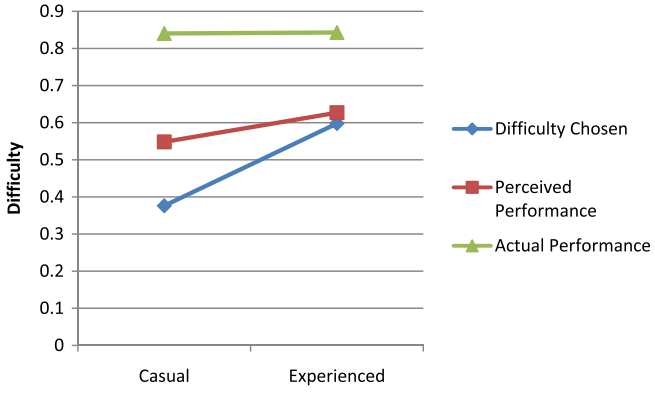


Fig. 4. Comparison of perceived performance and actual performance between casual and experienced players (Method A).

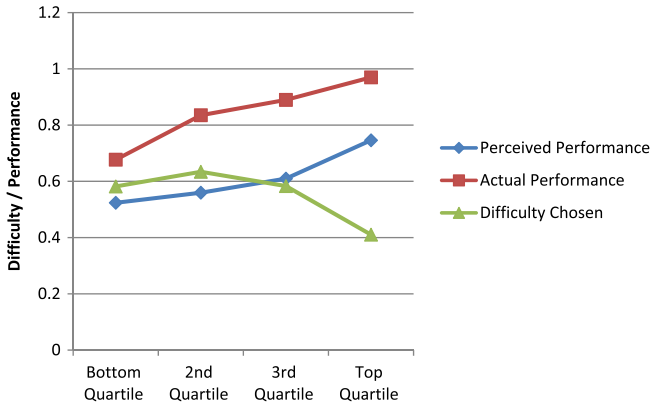


Fig. 5. Comparison of perceived performance and actual performance (Method A).

Both sets of players vastly underestimated their own actual performance, however experienced players demonstrated better self-assessment, since their perceived performance is closer to their actual performance, complimenting the Dunning-Kruger effect [12]. Please note that the actual performance of the average casual player and experienced player is only similar due to the method used (Method A) to calculate the actual performance of each player.

Instead of comparing casual and experienced players directly, Fig. 5 divides participants into quartiles depending on their actual performance using Method A, similar to how Dunning and Kruger [12] analysed their own results.

Those in the top quartile performed the best, because they chose an easier difficulty setting. Those in the bottom quartile however were the best at assessing their own performance, even though they performed the poorest (taking into account the difficulty they chose). Table 3 shows what types of players were within each quartile. Each quartile is closely balanced with similar amounts of casual and experienced players (Table 3).

It is very unreliable to use these results to compare them against the findings of Dunning and Kruger [12], since the performance of each participant was calculated using Method A. For example, a poor performer could choose the easiest difficulty and achieve the best score possible; whilst a better performer could choose the hardest difficulty, but attain an average performance. Whilst the poorer performer performed perfectly, it does not mean he or she is more skilful than the better performer.

Method B resolves this issue and is used to calculate the actual performance of each participant in Fig. 6.

Table 2
Minimum and maximum values used for the different difficulty parameters.

Parameter	Minimum	Maximum
Speed	2	15
Gate size	96	480
Distance between posts	400	800

Table 3
Number of participants in each quartile.

	Casual	Experienced
Bottom quartile	6	17
2nd quartile	4	18
3rd quartile	5	17
Top quartile	4	19

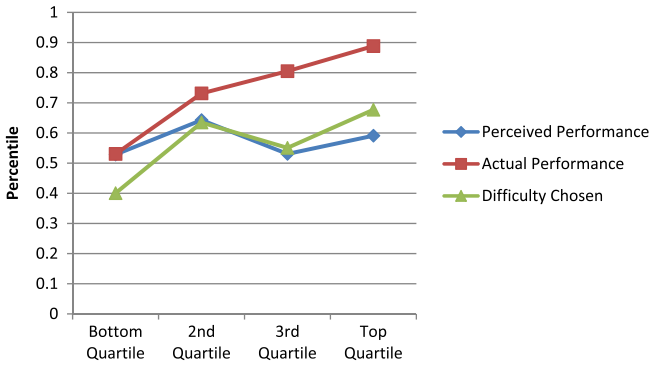


Fig. 6. Self-assessment in DDA mode of all players (Method B).

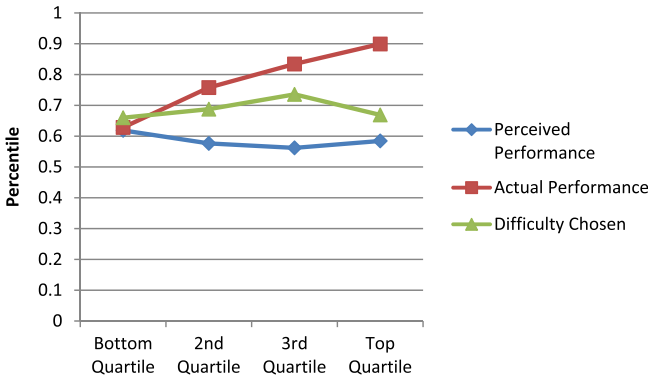


Fig. 8. Perceived performance and actual performance of experienced players in DDA mode (Method B).

Table 4
Dynamic difficulty mode participants.

	Casual	Experienced
Bottom quartile	7	4
2nd quartile	1	10
3rd quartile	3	8
Top quartile	1	11

Table 4 shows what types of players were in each quartile.

The bottom quartile consisted mostly of casual players, however the participants within this quartile interestingly demonstrated the best self-asssessment. The third and top quartile however, underestimated their performance. This does not support the Dunning-Kruger effect, since the bottom quartile did not overestimate their performance, nor did they show poor self-assessment. However, the better performers did underestimate their performance, which shares some similarity with the findings of Dunning and Kruger [12].

Next, the performances of both casual and experienced players are analysed separately. Fig. 7 compares the performance and perceived performance of casual players in DDA mode.

Casual players in the bottom quartile overestimated their performance greatly, whereas other quartiles underestimated their own performance. All quartiles however demonstrated negative motives, since they chose very easy difficulties that were not reflective of their ability.

In contrast to poor casual players, experienced players in the bottom quartile demonstrated the best self-assessment (Fig. 8). Similar to the casual players however, experienced players in the latter quartiles still underestimated their own performance. In both figures, the perceived performance seems to mirror the difficulty the player chose, which suggests the difficulty chosen influenced the player's perceived performance. In most instances (excluding the second quartile in Fig. 7), the easier the starting difficulty, the better the player thought they performed.

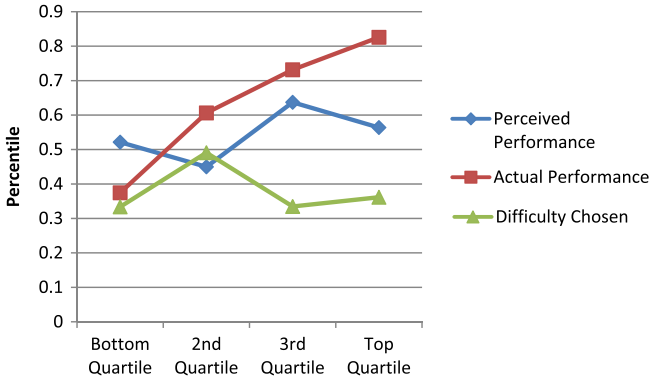


Fig. 7. Perceived performance and actual performance of casual players in DDA mode (Method B).

Table 5 compares the self-assessment between casual and experienced players. The bottom quartiles of both casual and experienced players differ greatly. Casual players who performed poorly overestimated their own ability by 14%, whereas experienced players, who performed poorly, closely matched their actual performance.

Whilst there are no significant correlations when comparing the results of casual players to experienced players; there is a noticeable pattern with the experienced participants. The experienced players underestimated their ability more so if they performed better, therefore their self-assessment worsened as their performance increased. Whilst this trend was not visible for the set of casual players, this may be due to the group's small sample size.

Contrary to the Dunning-Kruger effect, the trained (experienced) and better (top quartile) participants demonstrated the poorest self-assessment. A possible reason behind this is that the effect of DDA hindered their perceived performance value. In other words, once a player reached their performance threshold, they would have been unable to surpass it; therefore the difficulty would have decreased. Once the player reached their performance threshold again the same would have occurred. This situation would make the player feel that they are performing averagely, even when they actually are not, when considering the range of difficulty levels the game provides.

Each quartile chose a difficulty lower than their actual performance, besides the poorest experienced players. In an ideal scenario, each player would choose a difficulty that correlates with their ability; however in this instance there seems to be no correlation between the ability of the player and the difficulty chosen.

This demonstrates that most players do not choose a difficulty relating to their actual ability, but choose a difficulty relating to their experience. Experienced players for example, chose difficulties ranging from the medium-hard setting regardless of their ability, whilst casual players chose difficulties ranging from easy-medium, regardless of their ability also.

4.2. Comparison of skill and ability

Table 7 provides a direct comparison between different player types in the different difficulty modes to discover if they felt that the game's difficulty matched their skill.

Forty participants who played the game in DDA mode thought that the game's difficulty matched their skill, whilst five did not; suggesting that the DDA implementation was successful. Con-

Table 5
Comparison of self-assessment between casual and experienced players (Method B).

Performance	Casual				Experienced			
	Bottom ($n = 3$)	Second ($n = 3$)	Third ($n = 3$)	Top ($n = 4$)	Bottom ($n = 8$)	Second ($n = 8$)	Third ($n = 8$)	Top ($n = 8$)
Perceived (%)	52.13	44.97	63.70	56.38	61.88	57.64	56.20	58.45
Actual (%)	37.44	60.62	73.14	82.57	62.83	75.78	83.42	89.91
Difference	14.69*	−15.66	−9.44	−26.20	−0.95*	−18.14	−27.22	−31.46

versely, participants showed uncertainty as to whether the game matched their ability in static difficulty mode. Understandably, these results are influenced by the difficulty each participant chose, demonstrating that approximately 51.1% of participants thought that the difficulty they chose matched their skill.

The majority of experienced players thought that the game matched their skill in DDA mode. Twenty eight participants thought the game matched their skill, whilst four thought it did not. Unsurprisingly, the results for the static difficulty mode are different. Seventeen experienced players thought the game matched their skill, whilst 22 did not. This demonstrates that 43.5% of experienced players picked a difficulty that they thought matched their skill. Experienced players who thought that the game's difficulty matched their skill ($n = 45$) rated the game at a mean of 69%, whilst those who did not think the difficulty matched their skill ($n = 26$), rated the game at a mean of 60%. This does show that in this experiment, experienced players preferred if the game matched their skill, however the small 9% difference makes these findings quite insignificant.

One hundred percent of casual participants ($n = 6$) thought that the difficulty they chose in static difficulty mode matched their skill. Ninety two percent of casual participants ($n = 12$) thought that DDA mode matched their skill, whilst one participant did not. These results however may be unreliable due to the small sample size.

The results obtained from the static difficulty mode are quite obsure. As we already analysed in Table 6, the difficulties chosen by each group were not reflective of their ability, which creates an uncertainty as to why many players assumed that the difficulty they chose matched their ability. We suggest that the DDA mode miscalibrated their actual ability, due to a recurring cycle of reaching their ability peak.

4.3. Player enjoyment

Participants who felt that they performed better, rated the game higher (Fig. 9). In addition, participants who chose easier difficulties enjoyed the game more so.

Casual players rated the game higher if in DDA mode, whereas the experienced players preferred the static difficulty mode (Fig. 11). However, the differences between the ratings are too small to suggest that one group of players have a particular preference over another. Understandably, both sets of players found the DDA mode harder. As Fig. 10 showed, the lower the perceived difficulty, the higher the perceived performance – a higher perceived performance increased the rating of the game across both groups of players and difficulty modes.

Table 7
Percentage of players who thought the difficulty matched their skill.

	Static difficulty	DDA
All	51.1% (23:22)	88.8% (40:5)
Casual	100% (6:0)	92% (12:1)
Experienced	43.5% (17:22)	87.5% (28:4)

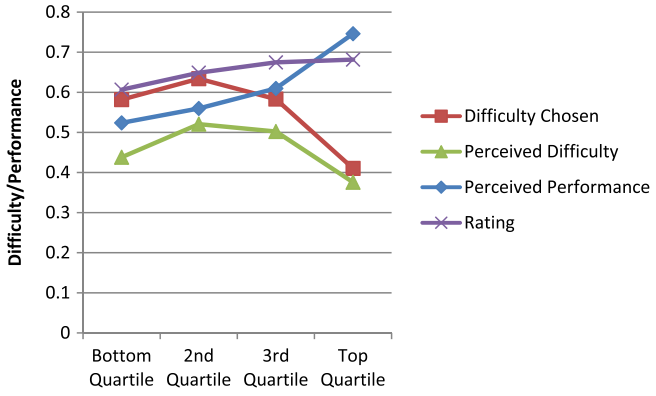


Fig. 9. Casual and experienced players' ratings in both difficulty modes (Method A).

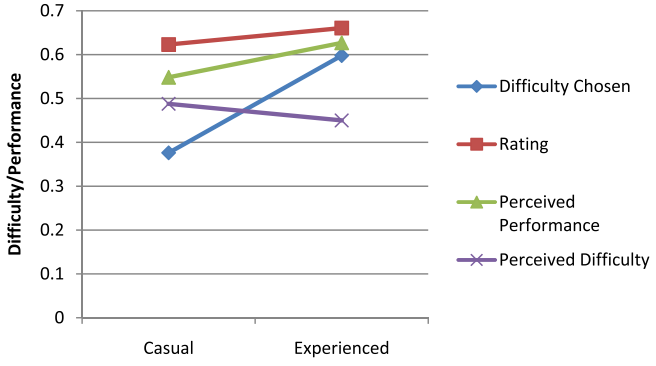


Fig. 10. Comparison of ratings between casual and experienced players in both difficulty modes (Method A).

Casual players who performed the poorest seemed to enjoy the game much more than their counterparts (Fig. 12). Whilst they did perform the poorest, they also found the game slightly easier than

Table 6
Comparison of difficulties chosen to the player's actual performance (Method B).

	Casual				Experienced			
	Bottom	Second	Third	Top	Bottom	Second	Third	Top
Chosen difficulty (%)	33.37	49.07	33.47	36.15	65.99	68.76	73.56	66.89
Actual performance (%)	37.44	60.62	73.14	82.57	62.83	75.78	83.42	89.91
Difference	−4.08	−11.56	−39.68	−46.42	3.16	−7.01	−9.86	−23.03

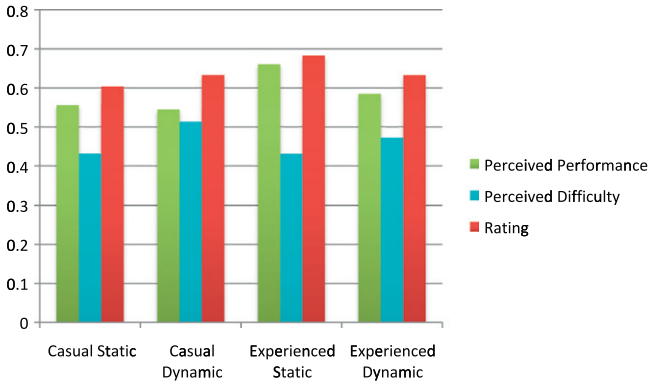


Fig. 11. Comparison of casual and experienced player's ratings in different difficulty modes.

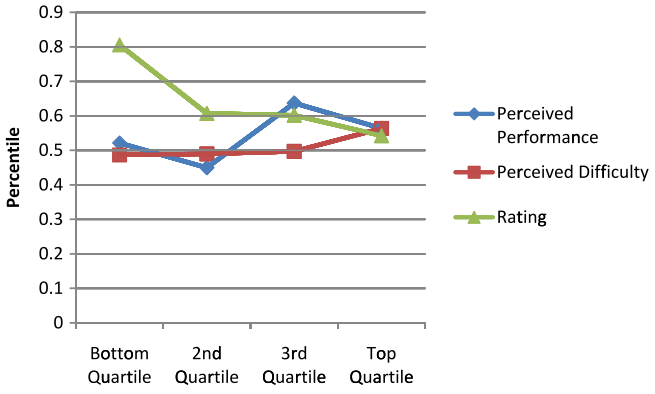


Fig. 12. Casual player ratings in DDA mode (Method B).

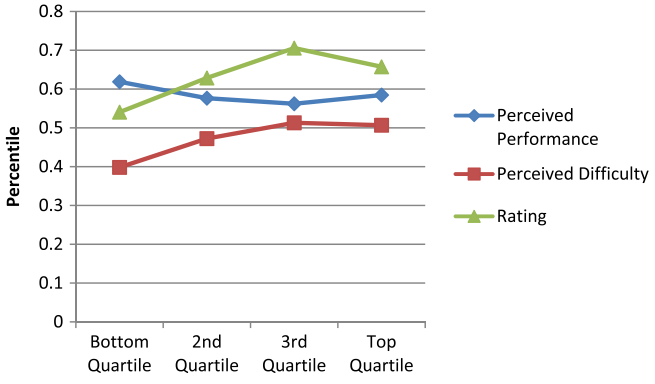


Fig. 13. Experienced player ratings in DDA mode (Method B).

other participants. On the other hand, the top quartile scored the game the lowest and found the game to be more difficult than the other players. These findings indicate that casual players enjoy a game more so if it is easier, therefore play for fun and relaxation, and not necessarily for the challenge.

Casual players who chose a high difficulty initially would have entered the game close to their peak difficulty; therefore would have been challenged to the highest of their ability consecutive times, increasing their frustration. Casual players who chose lower difficulties however, would have had more time to adjust and master the game, before reaching their peak difficulty, which also supports the use of the up and down difficulty direction [7].

In Fig. 13, the experienced players in the third quartile enjoyed the game the most, whereas the bottom quartile enjoyed the game the least, mirroring the effect seen from casual players in the bottom quartile (Fig. 12).

The third quartile rated the game higher than other quartiles, however found the game the most difficult and also thought that they performed the poorest. There is a visible trend between the perceived difficulty and the rating given by the experienced players. They both share the same trend, which suggests that experienced players seem to enjoy the game in relation to the challenge the game provides.

5. Conclusion

5.1. Research conclusion

We aimed to investigate the effects of game difficulty on player enjoyment, to help discover what game difficulty different types of players want. Hypothesis 1 was partially supported by our findings, since a quartile of the experienced players demonstrated the best self-assessment; however this quartile consisted of the poorest of performers, suggesting that the DDA mode miscalibrated the participants' perceived performances. Expectedly, Hypothesis 2 was fully supported. Casual players did choose easier difficulties regardless of their actual ability, but perhaps not just because their player profile portrays negative motives. It appeared that casual players enjoyed the easier difficulties, therefore chose the easier difficulties regardless of their ability. Hypothesis 3 was also fully supported; since experienced players enjoyed the game in relation to the challenge provided.

The key recommendation based on our findings is to ensure that the target audience's desires are met – not the designer's. A game's difficulty should not just be matched to the highest ability of the player continually, but be tweaked for the player's past gaming experience. As mentioned, the designer needs to consider why the player wants to play. Is it for relaxation? Is it for the challenge? We demonstrated the core differences between casual and experienced players, to help illustrate the best practices to use when designing difficulty management systems for different target audiences.

The following are concluding remarks to consider, when managing game difficulty:

5.1.1. Recommend a difficulty for the player based on their play style

As we discovered, both casual and experienced players chose difficulty settings that were not suited to their play style. For example, an experienced player who chooses a medium difficulty only to find it too easy may become quickly bored. Monitor the performance of the player to recommend a higher or lower difficulty for them, depending on their previous performance(s) and experience.

5.1.2. Let the player know how well they are doing

Both casual and experienced players demonstrated poor self-assessment. Ensure that the player knows how well they are performing. If they are performing well, it may give them a physiological boost. If they are performing poorly it may increase their motivation to rise to the challenge, or convince them to move to a difficulty setting that they would enjoy more.

5.1.3. Ensure that the game's learning curve is representative of the game's target audience

Casual gamers in this experiment needed more time to adjust to the game, therefore the learning curve needs to be reflective of its

target audience. An experienced player on the other hand, would likely require less time to adjust to the challenge the game offers.

5.1.4. Ensure that the game's challenge meets the needs of the player – not their ability

Experienced players demonstrated that they enjoy the game more so if there is a challenge to be met, whereas casual players demonstrated that they enjoy easier levels. Balance the pacing and difficulty of the game to meet the needs of the target audience.

5.1.5. Test, test, test

Finally, the best way to ensure that the game's difficulty is suited to the target audience is to get the target audience to play it and provide their opinion.

5.2. Research limitations

There were areas of our research that could be improved upon. Whilst the sample size was respectively good, there were not enough casual players who took part in our research. A larger amount of casual players would evidently make the findings of our research more reliable.

As stated in Section 3, participants were able to choose their own difficulty. Since the difficulty influenced the possible amount of points the player could attain, this demonstrated an inconsistency when comparing the performance of different players; especially when using the results obtained from participants who played in the static difficulty mode. This meant the performances calculated would have only been reliable if the participants chose the same starting difficulty in static difficulty mode. Fortunately, the results obtained from participants who played the game in DDA mode resolved this problem, however in doing so, the sample size was halved which meant the reliability in results decreased.

No mechanisms were implemented to prevent players from continually playing the game. However, the amount of times a player played the game in the same gameplay session was recorded. This was originally done to monitor if and why a player would attempt to play the game again in the same session. For example, an experienced player may have more desire to better their score than a casual player, therefore play again. It is possible that players may have returned at a later time to play the game again and the game and because of this, the game may inaccurately assume that it is the player's first gameplay session. Consequently it is conceived to be unfair, if players have had multiple gameplay sessions recorded, since their skill levels would have likely increased in each gameplay session.

Players may have also inaccurately chosen their wrong classification as either a casual or experienced player. Whilst there are many factors that contribute to a player's classification (e.g. the types of games they play, the reason they play and the amount of time they spend on games), it may have been best to guide the player in choosing the most ideal classification, by providing a description of the typical characteristics a casual and experienced player portrays. The results obtained however do not show anomalies in the casual or experienced groups, suggesting that players selected their correct classification.

5.3. Future research

The findings of our research suggest that it would be best to use static difficulty options that are reflective of a player's experience (i.e. casual or experienced), rather than ability (i.e. easy, medium or hard). It would be interesting to design an experimental game with casual and experienced static options; exhibiting the preferences each group has, for example:

- Experienced Mode: Provides a greater challenge and increases the game's challenge at a medium rate.
- Casual Mode: Starts from an easier difficulty and slowly increases the game's challenge.

The enjoyment levels of each participant could be recorded from this experiment and compared to the exact same game utilising the common static difficulty options (i.e. easy, medium and hard), to help identify what the best way of producing static difficulty options is.

Whilst we outlined the preferences that casual and experienced players had, there is an uncertainty that the same would occur when tweaking in-game elements such as:

- The type of game (by genre)
- The control scheme (e.g. dance mat, Kinect, Wiimote)
- Introducing competitive elements (i.e. multiplayer, co-operative or high-scores)
- The game's duration

In particular, introducing competitive elements may increase or decrease a game's enjoyment or replay value, depending on the type of player. Future research could analyse the behaviour of both casual and experienced players in competitive situations:

- Which group aspires to attain the best score (by recording the number of times participants play the game)
- Whose skill increases at a better rate when playing multiple times
- When do both groups play (i.e. if there is a new high score does the player instantly play and attempt to beat the score or does the player play for the game's enjoyment?)

This would help establish how to manage multiplayer or high-scores within a game, depending on the target audience.

As mentioned previously, monitoring the impact that the game's duration has on casual and experienced players within different difficulty methods could provide interesting results. For example, will DDA help to improve a player's concentration, performance and enjoyment over longer periods of time, in comparison with static methods?

Our experiment could also be carried out in different environments, such as public tournaments or in private at home. The element of peer pressure in a public environment may increase or decrease the difficulty the player chooses unintentionally and/or alter their perceived performance, so the change in environment would hopefully provide interesting results.

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